RA-HandRisk

AI-Based Early Risk Screening

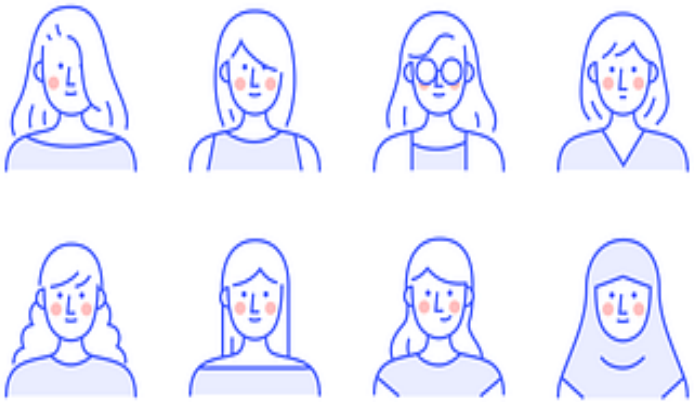
María José Viveros

Problem

- Early rheumatoid arthritis detection is difficult
- Symptoms are mild, gradual, and often ignored
- Requires clinical tests and specialists
- Example: morning stiffness mistaken as fatigue
- Delays lead to irreversible joint damage
- Need for early screening tools

Why It Matters

- 18 million people affected globally (WHO)
- Higher prevalence in women and older adults
- Early detection improves outcomes
- Reduces long-term disability and costs



Project Proposal

- Conceptual AI system (RA-HandRisk)
- Combines survey + hand images
- Goal: early risk identification
- Supports—not replaces—medical diagnosis

RA-HandRisk

Early RA Risk Detection with AI

Input

Survey +
+ Hand Images

Process

AI-based
Analysis

Output

Early Risk
Identification

Supports clinical decisions

Risk Factors

Biological

- Age
- Sex
- Genetics

Environmental

- Smoking
- Exposure

Clinical

- Stiffness
- Pain
- Fatigue

Visual

- Swelling
- Symmetry
- Alignment

Risk is the combination of all factors

System Input

Survey data: demographics + symptoms

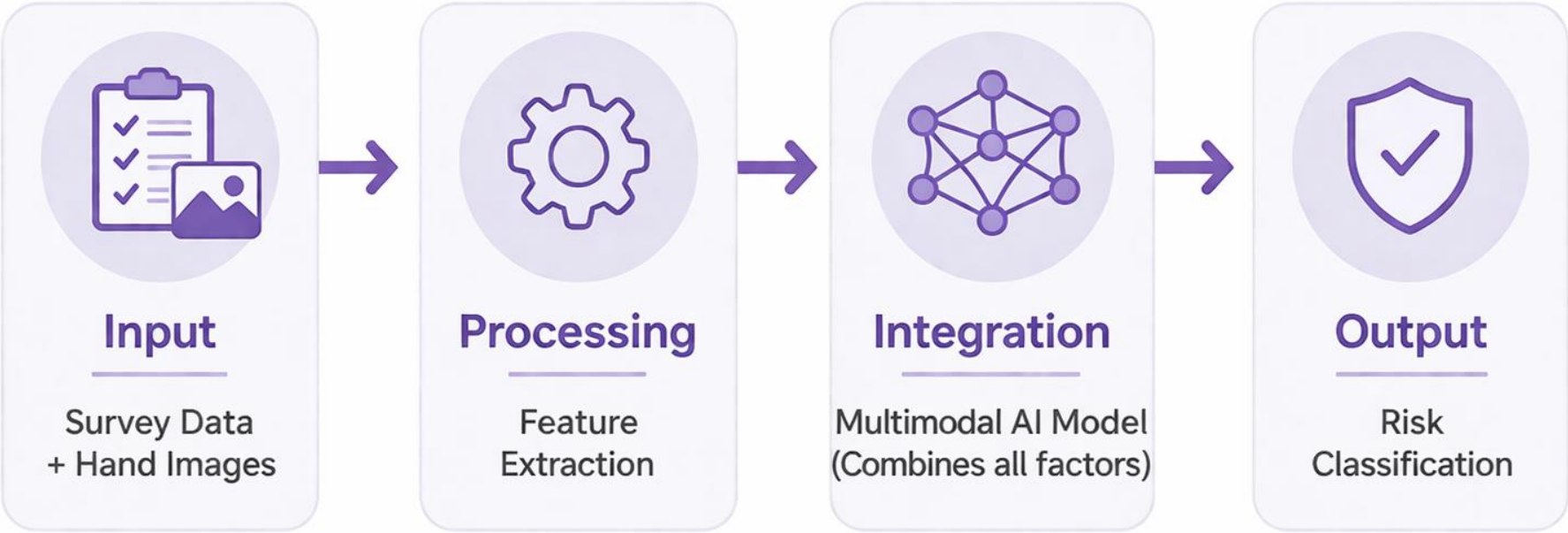
Examples: stiffness duration, pain, smoking

Image data: open hand + closed fist

Captures structure + movement limitations

System Flow (Figure 1)

RA-HandRisk system pipeline from data input to risk classification.



Survey + Images → Feature Extraction → AI Model → Risk Output

Detailed Solution Plan

1. Data collection
2. Data processing
3. Feature extraction
4. Risk integration
5. Output generation

Processing Stage

Survey → structured variables

Images → feature extraction

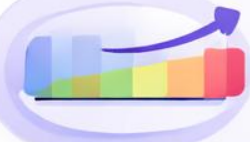
Detect swelling, symmetry, alignment

Example: fist closure = stiffness indicator

Risk Model

Probabilistic model (logistic concept)

Combines all variables



Outputs probability (0-1)

Converted into risk categories

Output

Low / Moderate / High risk

Explanation of contributing factors

Example: stiffness + smoking → higher risk

Recommendation for medical evaluation

Testing Plan

- Dataset validation
- Input quality control
- Survey and image testing
- Model evaluation
- Fairness and usability testing

Expected Results

Sensitivity > 85%

Specificity ~70-80%

AUC > 0.80

High usability (> 90%)

Low bias across groups

Ethical Considerations

Not a diagnostic tool

Requires medical confirmation

Protect user data

Avoid algorithm bias

Limitations

Images alone are not sufficient

Early RA may not be visible

Depends on dataset quality

Possible false positives/negatives

Limitations

Images alone are not sufficient

Early RA may not be visible

Depends on dataset quality

Possible false positives/negatives